

Muhammad Shaheer | FA24-BSE-104(B)

Object Oriented Programming

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CLO-03: Create a program using standard libraries.

7 December 2025

ASSIGNMENT#04

1. TO CREATE A HOME AUTOMATION GUI WITH THE FOLLOWING FUNCTIONALITY:

->TURN LIGHTS AND FANS ON OR OFF USING BUTTONS.

->ADJUST AIR CONDITIONER TEMPERATURE USING A SLIDER

```
import javax.swing.*;
import java.awt.*;

public class GUI {
    private JFrame frame;

    public GUI() {
        initializeGUI();
    }

    private void initializeGUI() {
        //Frame
        frame = new JFrame("Home Automation System.");
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setSize(650, 400);
        frame.setLayout(new BorderLayout());

        // Light Panel
        JPanel lightPanel = new JPanel();
        lightPanel.setLayout(new FlowLayout());
        lightPanel.setPreferredSize(new Dimension(0, 100));
        //Light Label
        JLabel label1 = new JLabel("Lights:");
        //Buttons
        JButton lightBtnOn = new JButton("ON");
        lightBtnOn.setBackground(Color.WHITE);
        JButton lightBtnOff = new JButton("OFF");

        lightBtnOn.addActionListener(e -> {
            lightBtnOn.setBackground(Color.GREEN);
            lightBtnOn.setOpaque(true);
            lightBtnOff.setBackground(Color.WHITE);
        });
        lightBtnOff.addActionListener(e -> {
            lightBtnOff.setBackground(Color.RED);
            lightBtnOff.setOpaque(true);
            lightBtnOn.setBackground(Color.WHITE);
        });
        //Added to Panel
        lightPanel.add(label1);
```

```
lightPanel.add(lightBtnOn);
lightPanel.add(lightBtnOff);

//Fan Panel
JPanel FanPanel = new JPanel();
FanPanel.setLayout(new FlowLayout());
FanPanel.setPreferredSize(new Dimension(0, 100));
//Fan Label
JLabel label2 = new JLabel("Fan:");
JButton fanBtnOn = new JButton("ON");
JButton fanBtnOff = new JButton("OFF");
fanBtnOn.addActionListener(e -> {
    fanBtnOn.setBackground(Color.GREEN);
    fanBtnOn.setOpaque(true);
    fanBtnOff.setBackground(Color.WHITE);
});
fanBtnOff.addActionListener(e -> {
    fanBtnOff.setBackground(Color.RED);
    fanBtnOff.setOpaque(true);
    fanBtnOn.setBackground(Color.WHITE);
});
//Added to Panel
FanPanel.add(label2);
FanPanel.add(fanBtnOn);
FanPanel.add(fanBtnOff);

// AC panel
JPanel AcPanel = new JPanel();
AcPanel.setLayout(new FlowLayout());
AcPanel.setPreferredSize(new Dimension(0, 120));
//Ac Label & Slider
JLabel label3 = new JLabel("AC Temperature:");
JSlider AcSlider = new JSlider();
AcSlider.setMinimum(16);
AcSlider.setMaximum(30);
AcSlider.setValue(26);

AcSlider.setMajorTickSpacing(2);
AcSlider.setMinorTickSpacing(1);

AcSlider.setPaintTicks(true);
AcSlider.setPaintLabels(true);

//value on right side
JLabel tempValue = new JLabel(AcSlider.getValue() + "°C");
AcSlider.addChangeListener(e -> {
    tempValue.setText(AcSlider.getValue() + "°C");
});
```

```

AcPanel.add(label3);
AcPanel.add(AcSlider);
AcPanel.add(tempValue);

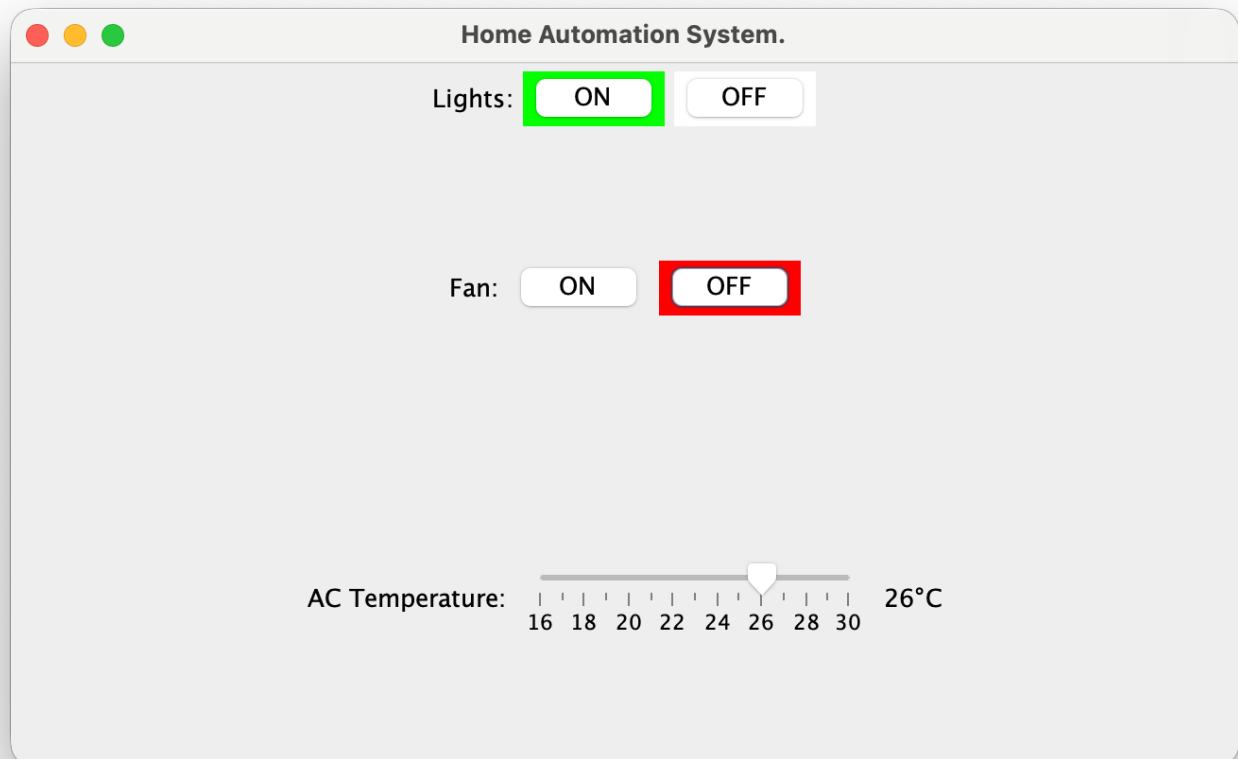
//Assembling Everything
frame.add(lightPanel, BorderLayout.NORTH);
frame.add(FanPanel, BorderLayout.CENTER);
frame.add(AcPanel, BorderLayout.SOUTH);

frame.setVisible(true);
}

public static void main(String[] args) {
    new GUI();
}
}

```

OUTPUT



2. WRITE THE CODE TO CREATE THE FOLLOWING GUI. THE GUI SHOULD HAVE FUNCTIONALITY OF CALCULATOR FOR PERFORMING ALL THE MATHEMATICAL OPERATIONS SHOWN IN THE GUI.

```
import javax.swing.*;
import java.awt.*;

public class Calculator {
    private JFrame frame;
    private JTextField input;
    private double num1 = 0;
    private String op = "";
    private boolean newNum = true;

    public Calculator() {
        frame = new JFrame("Calculator");
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setSize(300, 400);
        frame.setLayout(new BorderLayout());

        input = new JTextField();
        input.setPreferredSize(new Dimension(300, 40));
        frame.add(input, BorderLayout.NORTH);

        JPanel panel = new JPanel(new GridLayout(4, 4));

        String[] buttons = {
            "7", "8", "9", "/",
            "4", "5", "6", "*",
            "1", "2", "3", "-",
            "0", ".", "=", "+"
        };

        for (String text : buttons) {
            JButton btn = new JButton(text);
            btn.addActionListener(e -> {
                String cmd = e.getActionCommand();

                if ("0123456789".contains(cmd)) {
                    if (newNum) {
                        input.setText(cmd);
                        newNum = false;
                    } else {
                        input.setText(input.getText() + cmd);
                    }
                }
                else if ("+-*/".contains(cmd)) {
                    num1 = Double.parseDouble(input.getText());
                }
            });
        }
    }
}
```

```

        op = cmd;
        newNum = true;
    }
    else if (cmd.equals("=")) {
        double num2 =
Double.parseDouble(input.getText());
        double ans = 0;

        if (op.equals("+")) ans = num1 + num2;
        else if (op.equals("-")) ans = num1 - num2;
        else if (op.equals("*")) ans = num1 * num2;
        else if (op.equals("/")) ans = num1 / num2;

        input.setText(String.valueOf(ans));
        newNum = true;
    }
    });
    panel.add(btn);
}

frame.add(panel, BorderLayout.CENTER);
frame.setVisible(true);
}

public static void main(String[] args) {
    new Calculator();
}
}

```

